Adaptive Deformation-Learning and Multiscale-Integrated Network for Remote Sensing Object Detection

Xiyu Zhong, Jialei Zhan, Yuhang Xie, Lingtao Zhang, Guoxiong Zhou, Mingyue Liang, Kaitai Yang, Zonghao Guo, and Liujun Li

***Abstract*— Modern human productivity and daily life rely on identifying ground objects using remote sensing images (RSIs). Traditional remote sensing object detection (RSOD) techniques lack timeliness and accuracy and fail to meet practical demands. Existing deep learning algorithms face continued challenges when processing RSIs because of the diverse shapes and extensive scale variations of objects, of which a significant proportion are small scale. To address these challenges, we propose the PSWP-DETR, a transformer-based network that leverages adaptive deformation learning and multiscale integration for enhanced object detection in remote sensing. First, we propose PradatorConv (PdConv) to address the significant shape changes of objects because it adaptively learns the horizontal and vertical deformations to perceive the complex geometric features of RSIs. Second, we propose scale-wise differential modules (SDMs), which comprise multiscale convolution (MSC) and edge captor convolution (ECC). SDM integrates features across various scales and captures edge characteristics and local textures. This is advantageous for detecting multiscale objects, tiny objects with limited feature information. Finally, we propose the whale particle optimization (WPO) algorithm for learning rate optimization, which improves convergence speed and accuracy. Experiments using the VisDrone2019-DET, DIOR, and AI-TOD datasets demonstrated that PSWP-DETR achieves the best accuracy benefits, offering significant insights for future RSOD efforts. The source code will be available at https://github.com/Get1star/PSWP-DETR.git.**

***Index Terms*— Local texture features, offset convolution, remote sensing object detection (RSOD), transformer-based object detector, whale particle optimization (WPO).**

Received 2 September 2024; revised 15 November 2024 and 11 January 2025; accepted 10 February 2025. Date of publication 14 February 2025; date of current version 27 February 2025. This work was supported by the Hunan Provincial Natural Science Foundation Project under Grant 2025JJ50385. *(Xiyu Zhong and Jialei Zhan contributed equally to this work.) (Corresponding authors: Lingtao Zhang; Guoxiong Zhou.)*

Xiyu Zhong and Mingyue Liang are with the College of Computer and Mathematics, Central South University of Forestry and Technology, Changsha 410004, China (e-mail: xyzhong2023@163.com; yue19892853443@126.com).

Jialei Zhan is with the College of Systems Engineering, National University of Defense Technology, Changsha 410073, China (e-mail: jialeiz@163.com).

Yuhang Xie is with the School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: yhxie2022@163.com).

Lingtao Zhang, Guoxiong Zhou, Kaitai Yang, and Zonghao Guo are with the College of Electronic Information and Physics, Central South University of Forestry and Technology, Changsha 410004, China (e-mail: lzhangcsuft@163.com; zhougx01@163.com; yangkaitai@csuft.edu.cn; 1104743187@qq.com).

Liujun Li is with the Department of Soil and Water Systems, University of Idaho, Moscow, ID 83844 USA (e-mail: liujunl@uidaho.edu).

Digital Object Identifier 10.1109/TGRS.2025.3541441

I. Introduction

AUTOMATICALLY interpreting remote sensing images (RSIs) is a critical task in remote sensing. Object detection, aimed at identifying and locating specific category objects within images, is widely applied in various remote sensing tasks such as urban planning, environmental monitoring, resource management, military operations, and disaster early warning systems [1], [2], [3], [4]. Accurate and swift remote sensing object detection (RSOD) technology is pivotal for human productivity and daily life. The rapid advancement of unmanned aerial vehicle technology, satellite exploration, and other Earth observation methods in recent years has significantly enhanced the volume and quality of remote sensing datasets [2], [5], [74], [75]. Consequently, employing RSOD for the intelligent interpretation and thorough analysis of RSIs has become increasingly critical and urgent.

To initially detect objects in RSIs, early algorithms commonly relied on template matching [6], [7] or artificial feature-based modeling [8], [9]. While these methods offered significant interpretability, they were limited in feature representation capacity and robustness, particularly in remote sensing tasks within complex environments. With the development of deep learning, neural networks have demonstrated superior semantic representation capabilities. This has greatly facilitated innovative developments in various remote sensing domains such as hyperspectral remote sensing [10], [11], [12], infrared remote sensing [13], optical remote sensing [14], [15], [16], [17], [18], [19], and multimodal data remote sensing [20]. For example, in hyperspectral remote sensing, Li et al. [10] combined low-rank model approaches' with data-driven deep learning techniques to propose the LRR-Net for hyperspectral anomaly detection, significantly enhancing detection performance. In infrared remote sensing, the UIU-Net [13] embedded a small U-Net within a larger U-Net backbone, achieving multilevel and multiscale object representation learning. Notably, algorithms based on convolutional neural networks (CNNs), including two-stage algorithms like the R-CNN series [21], [22] and one-stage algorithms such as the YOLO series [23], [24], [25], have gained prominence in object detection. CNN-based remote sensing object detectors have shown remarkable progress in enhancing accuracy and real-time performance over traditional methods.

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

Unlike CNNs, the transformer [26] captures long-distance dependencies, providing a significant advantage in extracting information from images. DETR [27] innovatively integrates the transformer architecture into object detection tasks, leveraging CNNs as the backbone for extracting features from raw pixel values. This approach eliminates the need for numerous manually designed components, such as nonmaximum suppression and anchor generation, thereby enabling an end-to-end object detection method. Subsequently, a growing number of notable object detection works have adopted the transformer-based model [28], [29], [30]. Combined with CNN's powerful image feature extraction capabilities, it has effectively strengthened the model's representation of both global information and local information. RT-DETR [31] introduced an efficient hybrid encoder, replacing the traditional transformer encoder. This design enables flexible adjustment of inference speed through the use of various decoder layers, eliminating the need for retraining. Nonetheless, the current body of work on transformer-based RSOD is limited and encounters several challenges.

1) As illustrated in Fig. 1(a), the geometric shapes and texture features of objects in RSIs exhibit significant variability. This variability poses challenges for standard convolutional kernels, as their fixed rectangular receptive fields struggle to effectively capture and perceive these diverse geometric features [32].

2) As illustrated in Fig. 1(b), the area captured by the remote sensing imagery encompasses a wide array of diverse objects. Objects belonging to different categories, as well as those within the same category, frequently exhibit certain variations in scale, with small objects often constituting a larger proportion. Coupled with the variability in lighting conditions and the complexity of backgrounds, this leads to decreased clarity and recognizability of the object edges [33]. These factors pose challenges for transformers with limited local perception capabilities.

3) As illustrated in Fig. 1(c), during the model training process, an excessively high learning rate can impede convergence, whereas an excessively low learning rate can significantly slow down the convergence rate [34]. Given that RSI data typically possess high-resolution and complex backgrounds, model training often becomes more challenging.

To address the challenge of significant shape variations between different targets in remote sensing scenarios, Xu et al. [35] proposed a transformer-based contextual information aggregation network for RSI change detection. This network uses a filter-based visual tokenizer to divide each temporal feature map into multiple visual semantic tokens. Then, it incorporates progressive sampling with a vision transformer, effectively eliminating irrelevant change interference. By leveraging the transformer encoder, the network captures compact spatiotemporal contextual information from the token set, enabling the exploration of shape variation information. Lv et al. [36] proposed a novel adaptive regional spectral–spatial feature for land cover classification using high spatial resolution RSIs. This approach combines Gaussian-weighting spectral features with area shape index features to bridge the relationship between the area and shape of the adaptive region surrounding each pixel. In contrast, previous methods have focused on the importance of target shapes but have given less attention to shape differences across targets of varying sizes in remote sensing scenarios.

To address the challenges posed by multiscale objects, Zhou et al. [37] proposed CLT-Det, which employs a feature refinement module (FRM) to alleviate the semantic gap caused by variations in scale, achieving competitive performance in detecting multiscale distributed remote sensing objects. To address the issue of poor detection performance on small objects due to the limited local perception capabilities of the transformer, DIAG-TR [38], which is built upon the transformer, dynamically and in a layered manner integrates local details into overarching representations, effectively accounting for the scale discrepancies of geospatial objects. Li et al. [39] introduced the MSSA mechanism that performs feature extraction tailored to the varying sizes of remote sensing objects, significantly improving the accuracy, especially for small objects. However, these algorithms have limitations in addressing the shape variability of remote sensing objects. When objects extend beyond the fixed-shaped receptive fields of convolutional kernels, crucial features may be overlooked, rendering small-scale object detection in RSIs particularly challenging.

In order to identify suitable hyperparameters, such as learning rate, to improve detection performance, optimization algorithms such as SGD [40], Adam [41], and Adagrad [42] are typically employed. However, these algorithms may encounter the following issues: SGD is highly sensitive to the learning rate and thus requires manual tuning, which can be challenging. Adam and Adagrad are designed to automatically adjust the learning rate for each parameter using gradient statistics. While they often simplify the learning rate tuning process and lead to faster convergence, in some cases, their generalization performance is significantly worse than that of SGD [43]. These issues can potentially affect the training effectiveness of the model. In recent years, some researchers have applied biological-inspired intelligent optimization algorithms to the remote sensing field with promising results [44], [45]. For instance, Ragab [44] applied the Mayfly optimization algorithm to optimize the model's training hyperparameters, which led to improvements in the performance of remote sensing scene image classification. Despite these advances, optimization algorithms specifically designed for RSOD remain scarce.

To address the issues mentioned above, we introduce RT-DETR into RSOD tasks and propose PSWP-DETR based on it. The innovative contributions of this study are summarized as follows.

1) This research introduces PredatorConv (PdConv), designed to adaptively learn horizontal and vertical deformation offsets based on the input feature, thereby adaptively focusing on the features of a wide variety of shapes. This technique facilitates the flexible recognition and comprehension of complex geometric features in remote sensing objects, enhancing detection algorithms' capability to discern various geometric shapes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

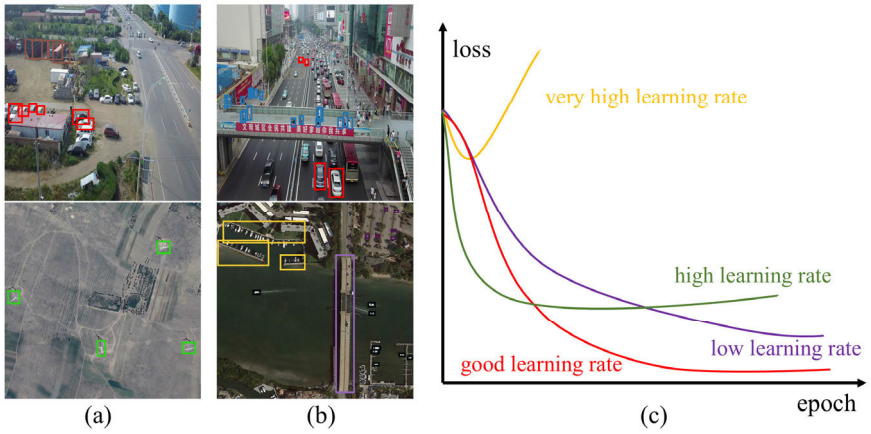


Fig. 1. Challenges in remote sensing object detection: (a) significant changes in the shape of the object due to inherent object characteristics, imaging angles, and occlusions; (b) wide range of scales between targets, with a high proportion of small targets; and (c) challenge of selecting an appropriate learning rate during model training.

2) This research introduces scale-wise differential modules (SDMs). SDM includes multiscale convolution (MSC) and edge captor convolution (ECC). ECC captures edge features and local textures using convolutional kernels tailored for specific edges, helping the model identify fine-grained details. By integrating multiscale features with these local details, SDM complements PdConv's ability to detect various shapes and sizes, enhancing overall detection accuracy.

3) This research introduces whale particle optimization (WPO) for learning rate optimization. WPO combines the global search capabilities of PSO with the local search strategies of WOA to achieve an effective balance between global and local search. This approach optimizes the learning rate adjustment process, accelerating model convergence and improving detection accuracy in RSOD.

4) We assessed our method using three public remote sensing datasets, comparing it against leading detectors to show PSWP-DETR's superiority. The experiments show that PSWP-DETR excels in identifying diverse object shapes and sizes, especially small objects common in RSIs.

II. Related Work

A. Efficient Remote Sensing Object Detection

With the continuous development of deep learning, increasingly deeper network architectures have enhanced model performance, but they have also introduced efficiency bottlenecks. In remote sensing applications, especially in scenarios requiring real-time processing, the real-time capability of deep learning models has become a critical consideration. To improve the efficiency of deep learning methods in remote sensing, researchers have proposed various optimization strategies, primarily focusing on model compression and network architecture design. Real-time object detection demands minimizing computational costs while maintaining accuracy. Among these strategies, model compression techniques are crucial for achieving real-time detection. To meet the requirements for low latency and high throughput, researchers have proposed several lightweight methods, including pruning, quantization, and knowledge distillation. For instance, Guo et al. [46] proposed a pruning method for RSI classification networks based on interpretable CNNs. First, an original interpretable CNN with a predetermined pruning rate was trained, and filters without interpretability were removed. Then, a sensitivity function was designed to evaluate the pruning risk of each layer's channels and to adaptively adjust the pruning rate for each layer. The pruning method based on the proposed sensitivity function proved to be effective, requiring minimal computational cost to search for redundant channels without compromising classification performance. Lu et al. [47] proposed an energy-based filter pruning framework to reduce the size of the original model. The energy of each layer is calculated by the proportion of eigenvalues below a specified truncation threshold, and filters are removed from the original layers based on the level of energy. The pruning method significantly accelerates network efficiency by eliminating unnecessary branches; however, it requires considerable manual design and has limited generalizability in remote sensing scenarios. Li et al. [48] applied knowledge distillation to remote sensing scenarios and proposed an instance-aware distillation (InsDist) method for efficient RSOD. InsDist combines feature- and relation-based distillation to effectively utilize instance-related information in the knowledge transfer from teacher to student. On the one hand, InsDist uses a parameter-free masking module to separate foreground instances related to the target from irrelevant background across multiscale features. On the other hand, InsDist builds relationships between different instances to enhance intraclass compactness and interclass dispersion, achieving significant results in handling complex RSIs. This knowledge distillation approach can be easily integrated into mainstream detectors with minimal computational overhead, making it a popular choice for model compression. Li et al. [49] applied a combination of quantization and hashing to large-scale RSI retrieval, proposing a quantized deep learning to hash framework. By binarizing the weights and activation functions into low-bit representations, this framework achieves retrieval with relatively minimal storage and computational resources. Zhan et al. [50] proposed a binarized RSOD network named HbiPiDiNet. First, they designed a binary

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

pixel difference convolution (BiPDC), which utilizes local binary patterns (LBPs) to capture local contrast information that is absent in traditional binary convolutions, thereby enhancing the representation of small objects. They then constructed a heterogeneous kernel fusion convolution block (HKFCB), which effectively represents multiscale LBP and BiPDC under multiscale binary convolutions, significantly improving the performance of state-of-the-art binary detection methods. Zhang et al. [51] combined quantization and distillation to design a structure called guided quantization self-distillation (GQSD). This innovative approach leverages the synergy between quantization and distillation to achieve a lightweight model. The training process of the quantized model is guided by a full-precision model, eliminating the need for a large pretrained model and saving substantial time and costs. However, the implementation of network quantization still requires considerable computational resources for training and necessitates corresponding hardware support, which limits its widespread adoption for real-time acceleration on GPU devices.

Since deep learning has been applied to object detection, and with the emergence of numerous CNN-based and transformer-based detectors, an increasing number of researchers have focused on designing efficient and real-time object detectors. A classic representative of CNN-based object detectors is the YOLO series [23], [24], [25]. Thanks to YOLO's real-time performance and its full-image prediction approach, many YOLO-based variants have been applied to RSOD. YOLO transforms object detection into a regression problem, enabling the detection of all object bounding boxes and classes in a single forward pass. Its full-image prediction approach allows the utilization of global image information for object localization, eliminating the need for a separate region proposal step as required by other methods. This gives YOLO strong object detection capabilities in RSIs, especially when object scales vary significantly. RSIs often contain vast background information as well as objects of various sizes. YOLOs are able to effectively extract information from global features and accurately detect objects at different scales. Zhu et al. [52] proposed TPH-YOLOv5, which adds an additional prediction head to YOLOv5 for detecting objects at different scales. They then replaced the original prediction head with a transformer-based prediction head to explore the predictive potential of self-attention mechanisms while also integrating the convolutional block attention model (CBAM) to identify attention regions in scenes with dense objects. Lin et al. [53] proposed YOLO-DA, a YOLO-based model with a decoupled attention head, which introduces minimal complexity. They designed an attention module at the end of the detector to guide the neural network in extracting more effective features from complex backgrounds while minimizing additional computational overhead and significantly improving detection performance. Zhang et al. [15] proposed FFCA-YOLO, which optimizes both accuracy and speed for airborne real-time processing by introducing three innovative, lightweight, and plug-and-play modules: feature enhancement module, feature fusion module, and spatial context aware module. These modules improve the network's ability to perceive local regions, fuse multiscale features, and model cross-channel and spatial global dependencies while minimizing added complexity. This enhances the weak feature representation of small objects and suppresses confusing background information. These contributions effectively explore the accuracy–speed tradeoff of YOLO-based networks in remote sensing scenarios. With the development of transformer models, the first real-time network based on transformer, RT-DETR [31], was proposed. RT-DETR designs an efficient hybrid encoder that quickly processes multiscale features by decoupling intrascale interaction and cross-scale fusion, thereby improving speed. In addition, it introduces the uncertainty-minimal query selection to provide the decoder with high-quality initial queries, thus enhancing accuracy. It outperforms previous state-of-the-art YOLO models in both speed and accuracy. The work of RT-DETR has also been applied to high-resolution RSOD tasks. For example, Wang et al. [54] proposed a real-time end-to-end high-resolution RSOD method based on RT-DETR. They first introduced a CGA-IFI for intrascale feature interaction, then proposed DRB-CFFM, which uses an expanded reparam block to promote cross-scale feature interaction. Finally, they enhanced the bounding box regression loss function with EIoU. This study focuses on overcoming the challenges posed by large shape differences of objects and the detection of multiscale objects, particularly small targets, in remote sensing scenarios. Based on RT-DETR, we propose our RSOD network, PSWP-DETR, which aims to improve the performance of RSOD.

B. Convolutional Design for Shape-Differentiated Objects

In RSIs, objects within the same category and across different categories often exhibit significant shape variations. Traditional CNNs employ convolutional kernels with fixed shapes and sizes, which limits the model's ability to capture complex geometric features. Designing convolutional weights based on the shape of the target object can help obtain features that better align with the target's geometry, thus facilitating the extraction of more relevant features. The famous deformable convolution [32] was introduced to address the inherent limitations of geometric transformations in CNNs, achieving outstanding performance in complex detection and classification tasks. Subsequently, several studies [55], [56], [57], [58] specifically designed convolutional kernels to adapt to the complex shape variations of objects in object detection tasks. For example, Kim et al. [55] introduced a dynamic adaptation mechanism in the convolutional kernel space, proposing a shape-adaptive kernel design that enhanced the ability of CNNs to accommodate shape variations in input features. Chen et al. [56] proposed an adaptive deformation network that enhances the ability of individual convolutions to focus on relevant image content by reconstructing deformable convolutions and ROI pooling. However, despite these methods partially mitigating the challenges posed by geometric shape variations, their ability to focus on local features within complex backgrounds remains insufficient. To overcome the difficulties in detecting and perceiving geometrically complex structures in images, Qi et al. [59] introduced a novel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

framework called DSCNet, which includes dynamic snake convolution (DSConv). DSConv enhances the network's ability to adaptively focus on fine-grained features, particularly targeting small and fragile local structures that are difficult to perceive. By incorporating morphological constraints specific to serpentine structures, DSConv improves the perception of these local features. In addition, DSConv has demonstrated promising performance in medical image processing tasks, providing important insights for our study. Building upon this, we propose PradatorConv (PdConv), which adaptively learns deformation offsets in the horizontal and vertical directions on the input feature maps, allowing it to capture features of objects with significant shape variations.

C. Multiscale Feature Representation for Object Detection

One of the main challenges in object detection is effectively representing objects at drastically different scales, and the varying object scales in RSIs present a significant challenge in remote sensing scenarios. Current object detectors typically utilize multiscale features to address this issue. As one of the pioneering works in multiscale feature representation, FPN [60] proposed a top–down path to combine multiscale features. PANet [61] further added a bottom–up path on top of FPN. NAS-FPN [62] and Auto-FPN [63] were proposed to automatically design cross-scale connections through neural architecture search. Tan et al. [64] introduced BiFPN, a simplified, repeated version of PANet. YOLO-MS [65] rethinks the multiscale feature representation learning in real-time object detectors by examining the use of different kernel sizes from both global and local perspectives. It constructs an encoder with strong multiscale feature extraction capabilities, significantly enhancing the multiscale feature representation in real-time object detection. With the advent of transformers, multiscale structures have also been incorporated into transformer-based detectors. Zhu et al. [28] introduced deformable DETR, where the proposed deformable attention module naturally aggregates multiscale feature maps through the attention mechanism, eliminating the need for a feature pyramid network. Although these methods account for multiscale features, they lack specialized designs tailored to the specific characteristics of remote sensing scenarios. In recent years, some studies have achieved promising results by designing multiscale structures specifically suited to the unique features of remote sensing targets. For example, Gao et al. [66] proposed a global multiscale fusion network without attention, utilizing a spatial bias module to capture long-range dependencies as part of the global information extraction module. This approach effectively captures global information and overcomes challenges posed by complex backgrounds. Zhan et al. [50] introduced HKFCB, which employs a multiscale structure to enhance the performance of binary convolution in remote sensing scenarios. We propose the SDM, which consists of MSC and ECC. The SDM captures features of multiscale objects through MSC and extracts features of blurred edges in small objects using ECC, thereby improving the detection accuracy of multiscale targets, especially small objects, in RSIs.

D. Bionic Algorithms Combined With Deep Learning

Biomimicry is widely applied in many modern scientific and technological fields. Biomimetic algorithms are a class of optimization algorithms inspired by biological organisms or systems in nature. By mimicking certain behaviors or processes in the natural world, these algorithms aim to solve complex optimization problems. Biomimetic algorithms include particle swarm optimization (PSO) [67], ant colony optimization (ACO) [68], sparrow search algorithm (SSA) [69], gray wolf optimizer (GWO) [70], and whale optimization algorithm (WOA) [71], among others. For example, the WOA [71] mimics the spiral hunting and encircling strategy of humpback whales, using random or optimal search agents, and demonstrates strong optimization capabilities. In recent years, the combination of biomimetic algorithms and deep learning has become an active research direction. Some researchers have achieved promising results on specific tasks by utilizing improved optimization algorithms [34], [72], [73]. For example, Dai et al. [72] combined the random exploration characteristic of the sparrow search algorithm and the gradual constraint mechanism of simulated annealing to propose the annealing-integrated sparrow optimization algorithm (AISOA). This approach helps the model avoid getting trapped in local optima while enhancing the recognition of blurred boundary features in rice disease images, maintaining stability during the training process. Liu et al. [34] proposed a reinforcement learning optimization algorithm based on particle swarm (ROPS). This innovative algorithm integrates PSO into reinforcement learning, enhancing the reward function and enabling the network to dynamically adjust the learning rate based on data characteristics, thereby improving the accuracy of building change detection. In summary, while biomimetic algorithms have found numerous applications in deep learning, their use in RSOD remains limited. To address this, we propose the WPO algorithm, which combines the powerful particle update mechanism of PSO with the diverse hunting strategies of WOA. This method helps optimize the learning rate during the RSOD model training process, thereby improving the model's detection performance.

III. Methodology

A. PSWP-DETR

We propose PSWP-DETR to address the challenges posed by significant shape and scale variations in objects within RSIs, which are due to factors such as imaging angles, distances, the inherent characteristics of the objects, and occlusions among them. This approach builds on the foundation of the RT-DETR-R50 (RT-DETR) object detector, specifically enhancing its performance for RSOD.

PSWP-DETR employs PdConv to fit and capture object features of various shapes in RSIs and then uses SDM consisting of MSC and ECC to merge semantic information and local texture information of objects at different scales. Finally, the WPO algorithm is used to optimize the learning rate, thereby enhancing detection accuracy. The undefined variables are described in detail in the subsequent subsections. The structure of PSWP-DETR is illustrated in Fig. 2. It uses the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

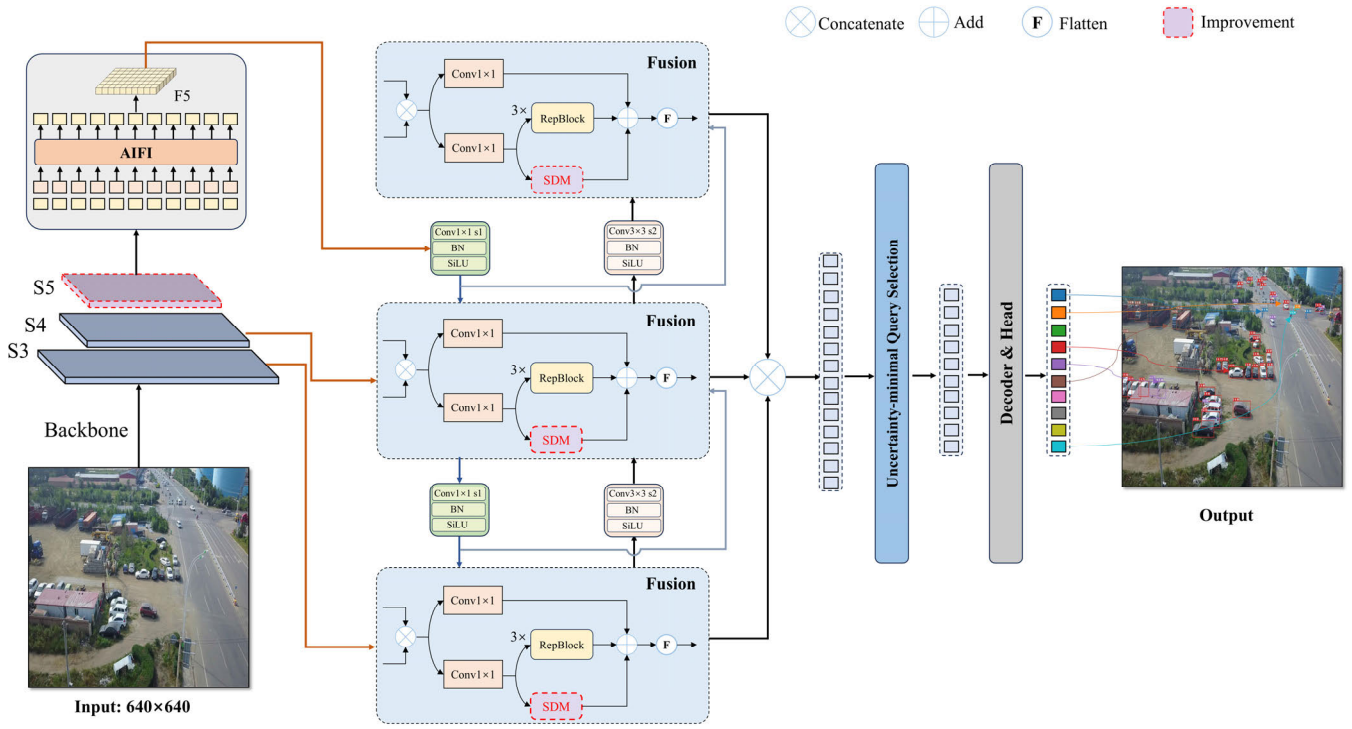


Fig. 2. Structure of PSWP-DETR.

output features from the last three stages of the backbone as inputs to the encoder. The efficient hybrid encoder transforms multiscale features into a sequence of image features through the attention-based intrascale feature interaction (AIFI) and the CNN-based cross-scale feature fusion (CCFF). Subsequently, the uncertainty-minimal query selection selects a fixed number of image features to serve as the initial object queries for the decoder. Finally, the decoder with auxiliary prediction heads iteratively refines the object queries to generate bounding boxes and confidence scores. As shown in Figs. 2 and 3, our improvements are primarily focused on the backbone and hybrid encoder. We utilize the proposed PdConv to enhance the stage S_5, in the backbone, and the proposed SDM to improve the fusion block in CCFF, with the SDM structure illustrated in Fig. 6. The fusion block is designed to fuse adjacent features into new features. This process can be formalized as follows:

$$\begin{aligned} Q = K = V &= \text{Flatten}(S_5) \\ F_5 &= \text{Reshape}(\text{Atten}(Q, K, V)) \\ \text{Output} &= \text{CCFF}(\{S_3, S_4, F_5\}) \end{aligned} \tag{1}$$

where Atten signifies the multihead self-attention and Reshape signifies restoring the shape of the feature to the same as S_5, which is the inverse operation of Flatten.

B. PredatorConv (PdConv)

Due to the characteristics of the objects themselves, the natural environment, observation angles, and limitations of remote sensing sensors, the shapes of objects may undergo significant changes. In response to this challenge and drawing inspiration from earlier studies, we propose PredatorConv

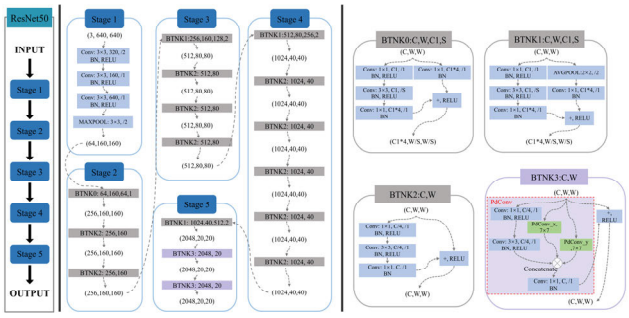


Fig. 3. Structure of the backbone.

(PdConv). On the one hand, PdConv employs 7 × 7 convolutional kernels that simulate the lateral (PdConv_x) and vertical (PdConv_y) movements of predators capturing prey in nature to capture the shape features of objects. On the other hand, it processes and extracts local, detailed features through 1 × 1 convolutional layers and 3 × 3 convolutional layers, achieving adaptability in detecting objects of different shapes in remote sensing scenarios. The working process of PdConv is illustrated in Fig. 4. For the offset convolutions part of PdConv, the lateral and vertical movements are achieved by stretching the standard convolutional kernel along the x-axis and y-axis, respectively, and then deforming with offsets to fit the shape of the object. Given the standard convolution coordinates as S, the central coordinate is $S_i = (x_i, y_i)$. A 3 × 3 kernel S with dilation 1 is expressed as

$$S = \{(x-1, y-1), (x-1, y), \ldots, (x+1, y+1)\}. \tag{2}$$

To prevent the receptive field from deviating from the object when the model autonomously learns offsets, we use

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

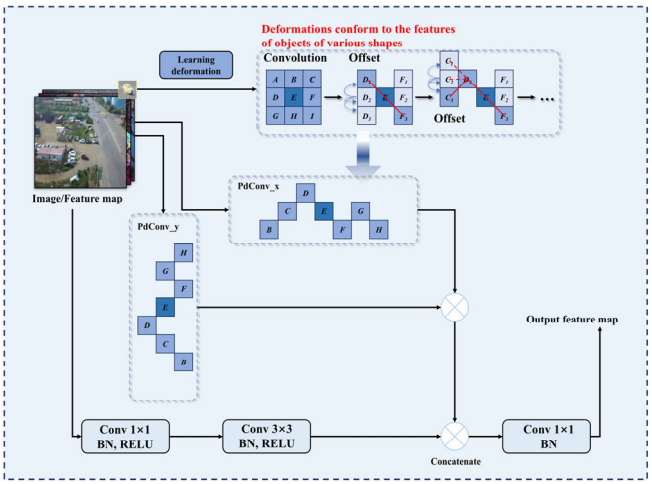


Fig. 4. Structure of PdConv.

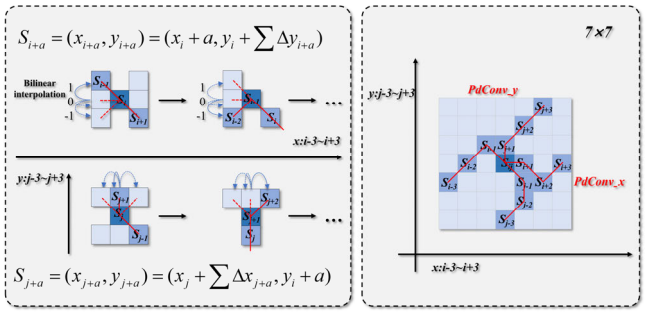


Fig. 5. (Left) Illustration of the coordinates of the offset convolutions in PdConv. (Right) Receptive field of the offset convolutions in PdConv.

an iterative strategy (see Fig. 5) to sequentially select the following observation positions for each object to be processed. This ensures the continuity of attention and prevents the receptive field from spreading too far due to significant deformation offsets. To make the convolutional kernel more flexibly focus on and perceive the complex geometric feature information of remote sensing objects, we employ a kernel size of 7, stretching the standard convolutional kernel separately along the x-axis and y-axis. Taking the x-axis as an example, the specific position of each grid in S is represented as $S_{i\pm a} = (x_{i\pm a}, y_{i\pm a})$, where $a = \{0, 1, 2, 3\}$ represents the horizontal distance to the center of the grids. Choosing each grid position $S_{i\pm a}$ within the convolution kernel S_i involves an accumulative process. Beginning at the central position S_i, the displacement from the central grid is determined by the position of the preceding grid: S_{i+1} is adjusted by an offset $\Delta = \{\delta|\delta \in [-1, 1]\}$ relative to S. Consequently, the offset must be summed, thereby guaranteeing that the convolution kernel adopts a linear morphological configuration. The values in the x-axis direction in Fig. 5 become

$$S_{i\pm a} = \begin{cases} (x_{i+a}, y_{i+a}) = \left(x_i + a, y_i + \sum_{i}^{i+a} \Delta y\right) \\ (x_{i-a}, y_{i-a}) = \left(x_i - a, y_i + \sum_{i-a}^{i} \Delta y\right). \end{cases} \tag{3}$$

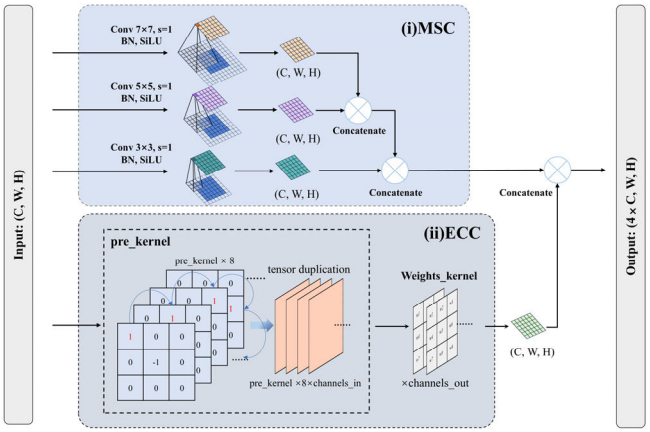


Fig. 6. Structure of SDM, which consists of MSC and ECC.

The values in the y-axis direction for (3) become

$$S_{j\pm a} = \begin{cases} (x_{j+a}, y_{j+a}) = \left(x_j + \sum_{j}^{j+a} \Delta x, y_j + a\right) \\ (x_{j-a}, y_{j-a}) = \left(x_j + \sum_{j-a}^{j} \Delta x, y_j - a\right). \end{cases} \tag{4}$$

The offset Δ is typically fractional, and therefore, bilinear interpolation is implemented as

$$S = \sum_{S'} B(S', S) \cdot S' \tag{5}$$

where S represents the fractional position in (3) and (4), S' enumerates all integer spatial positions, and B is a bilinear interpolation kernel divided into two 1-D kernels

$$B(S, S') = b(S_x, S'_x) \cdot b(S_y, S'_y). \tag{6}$$

As shown in Fig. 5, due to the 2-D (x-axis and y-axis) variation, during the deformation process, PdConv_x and PdConv_y can cover a 7 × 7 range to better perceive the key features of objects of various shapes in RSIs. In addition, we have incorporated a 1 × 1 convolutional layer and a 3 × 3 convolutional layer to extract local features from the images. After fusing these with the feature information output from the 2-D deformed convolution, we connect a 1 × 1 convolutional kernel for channel reduction.

PdConv is designed to adapt to the varying shapes of objects in RSIs, learning to deform based on the input feature map and focus adaptively on objects' diverse shapes. This enhances the model's capacity to recognize features from multiple directions, improving detection accuracy.

C. Scale-Wise Differential Modules (SDM)

Influenced by factors such as imaging distance and object characteristics, the scale of objects in RSIs exhibits significant variability and most display traits of small objects. Furthermore, variations in lighting and complex backgrounds reduce the clarity and recognizability of object edges. To address these

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

adverse effects, we draw inspiration from [76] and propose the SDMs, which consist of MSC and ECC. The structure is illustrated in Fig. 6.

First, MSC divides the feature map into $s(s = 1, 2, 3)$ scales, and the feature map $s(s = 1, 2, 3)$ is processed using convolution operations of size $k(k = 7, 5, 3)$. The feature map output from the first $s(s = 1)$ will be concatenated in sequence with the feature map output from the next $s(s = 2, 3)$, resulting in a comprehensive feature representation. After each convolution operation, the feature maps undergo normalization and activation through the sigmoid linear unit (SiLU) function, enhancing the feature's expressiveness and the network's training efficiency. Experimentally, as discussed in Section IV-D2, we find that the best performance of the SDM is achieved when the scale of MSC is set to $s = 3$.

Second, as shown in Fig. 6, ECC consists of eight predefined convolution kernels, each derived from a 9×8 matrix. The matrix includes rows representing distinct 3×3 kernels with binary patterns. The central element of each 3×3 kernel is set to -1, and one adjacent element is set to 1, with all other elements set to 0. This configuration allows each kernel to highlight local discontinuities and contrasts effectively. The matrix configuration is as follows:

$$\mathbf{pre_kernel}_i = \begin{bmatrix} \mathbf{1},\mathbf{0},\mathbf{0},\mathbf{0},-\mathbf{1},\mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0} \\ \mathbf{0},\mathbf{1},\mathbf{0},\mathbf{0},-\mathbf{1},\mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0} \\ \mathbf{0},\mathbf{0},\mathbf{1},\mathbf{0},-\mathbf{1},\mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0} \\ \mathbf{0},\mathbf{0},\mathbf{0},\mathbf{1},-\mathbf{1},\mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0} \\ \mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0},-\mathbf{1},\mathbf{1},\mathbf{0},\mathbf{0},\mathbf{0} \\ \mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0},-\mathbf{1},\mathbf{0},\mathbf{1},\mathbf{0},\mathbf{0} \\ \mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0},-\mathbf{1},\mathbf{0},\mathbf{0},\mathbf{1},\mathbf{0} \\ \mathbf{0},\mathbf{0},\mathbf{0},\mathbf{0},-\mathbf{1},\mathbf{0},\mathbf{0},\mathbf{0},\mathbf{1} \end{bmatrix}, (i = 1, 2, \ldots, 8). \tag{7}$$

Each 3×3 kernel is reshaped and duplicated to match the number of input channels, ensuring that each channel is independently processed to extract distinct features. Then, ECC processes the input data through two primary convolution stages.

First Convolution Stage: The input data are convolved with the pre_kernel. This stage captures the initial edge and texture features by applying the predefined binary patterns across all channels. The convolution is performed independently for each channel, ensuring that the features from each channel are enhanced based on the local pixel differences captured by the kernels.

Second Convolution Stage: Post initial feature extraction, a second convolution is applied using a set of trainable weights (weights_kernel). These weights are optimized during the training process to transform the initially highlighted features into more complex patterns. This complex transformation enhances the model's ability to discern subtle nuances in the remote sensing data, enabling precise detection of fine edges and intricate local features.

By integrating the precise feature enhancement of the pre_kernel with the adaptive learning capacity of the weights_kernel, the ECC specifically excels in detecting and classifying subtle edge distinctions and localized features crucial in remote sensing. This includes the identification of small objects, variations in natural and man-made environments, and the detection of targets obscured by natural phenomena.

Finally, SDM concatenates the output processed by ECC with the output processed by MSC to form the final output. A 1×1 convolution layer is then connected for channel reduction. SDM enables the model to effectively extract the object's multiscale features and local texture information through MSC and ECC, a characteristic that is highly beneficial for fine-grained recognition and efficient capture of edge texture features.

D. Whale Particle Optimization (WPO)

We propose WPO to optimize the learning rate during model training and enhance the performance of RSOD. By integrating the velocity and position update mechanisms of PSO [67] with the spiral predation strategy of WOA [71], WPO enhances both global and local search effectiveness. The procedure of the WPO is presented in Algorithm 1.

During the initialization phase, WPO begins by randomly initializing the positions of particles X_i^0 and their velocities V_i^0, within the search space defined by [lb,ub]. The historical best position for each particle is set to its current position: $P_i^0 \leftarrow X_i^0$, and the global best position P_{best} and global best fitness F_{best} are also established. The algorithm then enters the iterative process.

Algorithm 1 Whale Particle Optimization (WPO)

Input: dimension d, upper bound ub, lower bound lb, object function f, number of particles n, maxinum of iterations T, inertial weights w, individual cognitive factors $c1$, social cognitive factors $c2$
Output: global best position P_{best}, global best fitness F_{best}
Initialize
  For i from 1 to n:
    $X_i^0 \sim$ Uniform(lb,ub)
    $V_i^0 \sim$ Uniform(-0.1,0.1)
    $P_i^0 \leftarrow X_i^0$
  Determine $P_{best}$and F_{best}
Optimization Loop$t = 1$ **to** T
  Evaluate Fitness:
    For i from 1 to n:
      Compute $f(X_i^t)$
      Update P_i^{best} and F_i^{best} if $f(X_i^t)$ is better
      Update P_{best} and F_{best} if $f(X_i^t)$ is globally better
  Compute$a$**value:**
    $a^t = 2 - \frac{2t}{T}$
  Update Positions and Velocities:
    For i from 1 to n:
      Compute r, A^t, C^t, p
      Update X_i^{t+1} and V_i^{t+1} based on conditions
  Apply Bound Constraints:
    $X_i^{t+1} \leftarrow \text{clip}(X_i^{t+1}, lb, up)$
Return the Best Solution:
  Return P_{best}, F_{best}

1) The convergence coefficient a controls the transition of the search behavior from global exploration to local exploitation, dynamically decreasing linearly to 0 with the maximum

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

of iterations

$$a^t = 2 - 2 \cdot \frac{t}{\mathrm{T}} \tag{8}$$

where t is the current iteration number and T is the maximum number of iterations.

2) For each particle, WPO randomly generates a parameter r and performs specific computations to obtain coefficients A and C, as shown in the following equations:

$$A^t = 2 \cdot a^t \cdot r - a^t \tag{9}$$

$$C^t = 2 \cdot r. \tag{10}$$

Next, WPO implements a dynamic position update strategy based on a random probability p for each particle. This strategy adaptively adjusts the search range according to the current search status and historical information. Specifically, each particle moves closer to the best position of a randomly selected other particle j. When $p < 0.5 \& |A| > 1$, the particle's current position X_i^{t+1} is given by

$$D = |C^t \cdot P_j^{\text{best}} - X_i^t| \tag{11}$$

$$X_i^{t+1} = P_j^{\text{best}} - A^t \cdot D \tag{12}$$

where D is the distance calculation, P_{j}^{best} denotes the historical best position of particle j, and X_i^t represents the position of particle i at iteration t. When $p < 0.5 \& |A| \leq 1$, WPO gradually encircles the global optimal solution, conducting more in-depth exploration within the locally optimal regions of the solution space to update the current position X_i^{t+1}

$$D = |C^t \cdot P_{\text{best}} - X_i^t| \tag{13}$$

$$X_i^{t+1} = P_{\text{best}} - A^t \cdot D. \tag{14}$$

Conversely, when $p \geq 0.5$, WPO uses a spiral update method to update the particle's current position X_i^{t+1}, exploring in multiple directions around the global optimal solution, thereby increasing the chances of finding a better local optimum

$$D = |P_{\text{best}} - X_i^t| \tag{15}$$

$$X_i^{t+1} = D \cdot e^{\mathrm{b} \cdot l} \cdot \cos(2\pi l) + P_{\text{best}} \tag{16}$$

where b is a constant defining the spiral shape and l is a random factor influencing the spiral path.

3) Update the particles' velocity V_i^{t+1} and position X_i^{t+1}, and apply boundary constraints to X_i^{t+1}

$$V_i^{t+1} = \omega \cdot V_i^t + \mathrm{c}_1 \cdot \text{rand}() \cdot \left(P_i^{\text{best}} - X_i^t\right) + \mathrm{c}_2 \cdot \text{rand}() \cdot \left(P_{\text{best}} - X_i^t\right) \tag{17}$$

$$X_i^{t+1} = X_i^t + V_i^{t+1} \tag{18}$$

$$X_i^{t+1} = \text{clip}\left(X_i^{t+1}, \text{lb}, \text{ub}\right) \tag{19}$$

where ω is the inertia weight; c_1 and c_2 are cognitive and social coefficients, respectively; and rand() is a random number between 0 and 1.

4) Check if the current position is the particle's personal best. If it is, update the personal best position. Then, check if the current position is the global best. If it is, update the global best position.

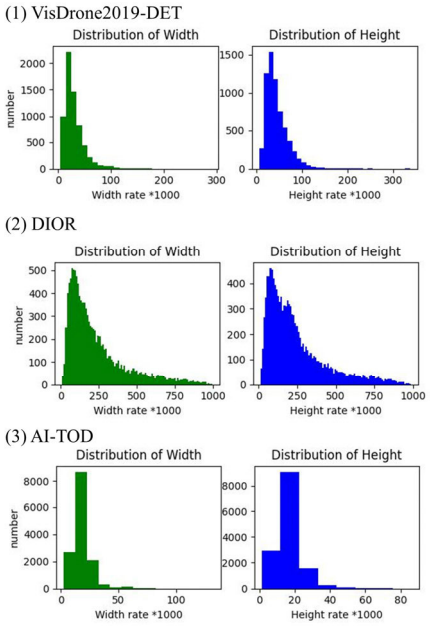


Fig. 7. Size analysis of training images of the datasets. (a) VisDrone2019-DET. (b) DIOR. (c) AI-TOD.

The WPO algorithm cleverly combines the hunting behavior simulation of the WOA with the information-sharing mechanism of PSO, effectively integrating global and local search strategies. This combination not only accelerates the broad exploration of the search space but also effectively avoids the problem of getting trapped in local optima through diversified update strategies and dynamic learning rate adjustments. In Section IV-D3, we conducted a detailed analysis of the comparative experiments for the WPO module, and the results further confirmed the superiority of WPO.

IV. Experiments

A. Datasets

Considering the characteristics of RSIs, we use VisDrone2019-DET [74], DIOR [2], and AI-TOD [75] as our experimental datasets. Fig. 7 displays their image size distribution, with average sizes of (1520, 1002) for VisDrone2019-DET, and (800, 800) for DIOR and AI-TOD. Median object-to-image size ratios are (0.025, 0.038), (0.17, 0.19), and (0.017, 0.016), showing that most objects occupy a small image portion.

1) VisDrone2019-DET: The VisDrone2019-DET dataset comprises 10 209 static images with high resolution. We randomly selected 6471 of these RSIs as the training set, 548 images as the validation set, and 1610 images as the test set. The image categories (ID) include pedestrian (0), people (1), bicycle (2), car (3), van (4), truck (2), tricycle (6), awning tricycle (7), bus (3), and motor (4).

2) DIOR: The DIOR dataset contains 23 463 images and 192 472 instances, covering a wide range of object scale variations. We randomly selected 11 725 images as the training set and 11 738 images as the validation and test sets. The image categories (ID) include airplane (1), airport (2), baseball field (3), basketball court (4), bridge (2), chimney (6), dam

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

TABLE I
HARDWARE AND SOFTWARE ENVIRONMENT

Hardware environment		Software environment	
CPU	56 vCPU Intel(R) Xeon(R) Gold 6330 CPU @ 2.00GHz	OS	LINUX
RAM	320GB	CUDA	11.2
Video memory	24 GB ×4	Python	3.8.17
GPU	RTX 3090 ×4	Paddlepaddle	2.5.0
Server platform	AutoDL	MMYOLO	0.6.0
		detrex	0.3.0

(7), expressway service area (3), expressway toll station (4), golf field (10), ground track field (5), harbor (6), overpass (7), ship (14), stadium (8), storage tank (9), tennis court (10), train station (11), vehicle (12), and windmill (20).

3) AI-TOD: The AI-TOD dataset contains 28 036 aerial images and 700 621 object instances, with the average size of the objects being approximately 12.8 pixels, which is much smaller than the objects in other datasets. We randomly selected 14 018 images as the training set and another 14 018 images as the validation and test sets. The image categories (ID) include airplane (1), bridge (2), storage tank (3), ship (4), swimming pool (2), vehicle (6), person (7), and wind mill (3).

B. Experimental Environment

All experiments were performed in identical hardware environments to minimize test result discrepancies from varying experimental setups and guarantee program stability and reproducibility. PSWP-DETR and RT-DETR were implemented using Paddlepaddle [77], while the YOLO series utilized MMYOLO [78] and the transformer-based series employed detrex [79]. The specific environments used are detailed in Table I.

We use RT-DETR pretrained on COCO2017 as the pretrained weights and the COCO AP metric for evaluation on single-scale input images. Given GPU limitations and training duration, the batch size was 2 for 72 epochs. The initial learning rate is 0.0005. The data augmentation includes random {color distort, expand, crop, flip, resize} operations, the benchmark of inference speeds, and the training strategy and hyperparameters of the decoder almost follow [31]. For the uncertainty-minimal query selection, we select the top 600 encoder features to initialize the decoder's object queries. The YOLO series and our models used 640 × 640 inputs, whereas the inputs for the transformer series have a short_edge_length of 800 and a max_size of 1333. The inference parameters were set with a score threshold of 0.5 and an intersection-over-union (IoU) threshold of 0.6 for detection box identification.

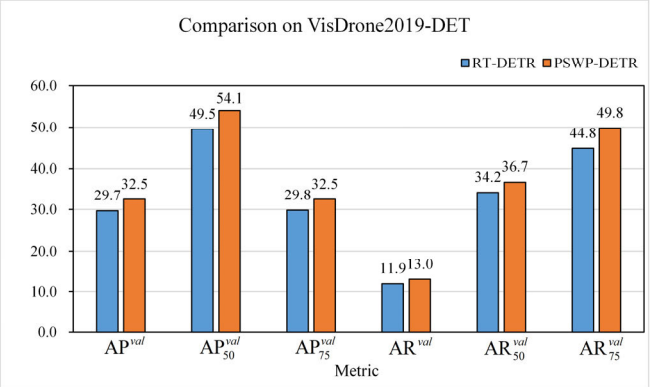


Fig. 8. Comparison on VisDrone2019-DET.

C. Performance Experimental Analysis

To thoroughly assess PSWP-DETR's performance, we conducted a comprehensive performance comparison on the VisDrone2019-DET dataset, calculating average precision (AP), average recall (AR), frames per second (FPS), parameters (Params), and GFLOPS. Fig. 8 shows that PSWP-DETR surpasses RT-DETR in both AP and AR across three IOU thresholds, with detailed data presented in Table II.

As shown in Table II, PSWP-DETR outperforms RT-DETR in accuracy across various IOU thresholds, excluding AP_L^{val}, with a notable 3.1% improvement in AP_S^{val}. Although AP_L^{val} decreased, we will address that in Section V. Nevertheless, on the DIOR dataset, our model excels at detecting objects of diverse scales, as evident in Table VII. PSWP-DETR also significantly outperforms RT-DETR in AR, highlighting its superior overall perception ability, especially for small objects with limited pixels. PSWP-DETR can capture subtle features more effectively. Although PSWP-DETR is slightly slower than RT-DETR in inference, its accuracy benefits are undeniable.

We evaluated PSWP-DETR across various remote sensing scenarios on the VisDrone2019-DET test set, with detections visualized in Fig. 9, which displays detection frames, categories, and confidence levels. RT-DETR missed occluded vehicles in group (a), and either confused or missed occluded objects in group (b), whereas PSWP-DETR correctly detected more occluded and feature-blurred objects. Under conditions of low-light and distant imaging angles, PSWP-DETR detected small targets with a lower pixel ratio more accurately than DETR and identified a greater number of targets in groups (c) and (d). This indicated that in various complex scenarios, PSWP-DETR outperformed RT-DETR in RSOD.

D. Validity Experiment of the Modules

In this section, we investigated the impact of each module's components on performance.

1) Effectiveness Experiment of Modules in PdConv: To assess PdConv's component effectiveness, we examined each part's impact, holding other variables constant. This included x-axis deformed convolution (PdConv_x), y-axis deformed convolution (PdConv_y), and kernel deformation parameters (Offset), with results in Table III. Findings showed that AP

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPREHENSIVE COMPARISON OF PSWP-DETR AND RT-DETR PERFORMANCE (THE BETTER VALUE IN THE TABLE WILL BE BOLDED)

Metrics	RT-DETR	PSWP-DETR	Metrics	RT-DETR	PSWP-DETR
AP^{val}	29.7	**32.5**	AR^{val}	11.9	**13.0**
AP^{val}_{50}	49.5	**54.1**	AR^{val}_{50}	34.2	**36.7**
AP^{val}_{75}	29.8	**32.5**	AR^{val}_{75}	44.8	**49.8**
AP^{val}_{S}	19.2	**22.3**	AR^{val}_{S}	34.5	**40.3**
AP^{val}_{M}	42.6	**45.4**	AR^{val}_{M}	59.2	**63.3**
AP^{val}_{L}	**69.1**	64.1	AR^{val}_{L}	79.4	**80.3**
FPS	**42.6**	32.8	Params(M)/GFLOPS(G)	41.48/68.19	98.89/140.86

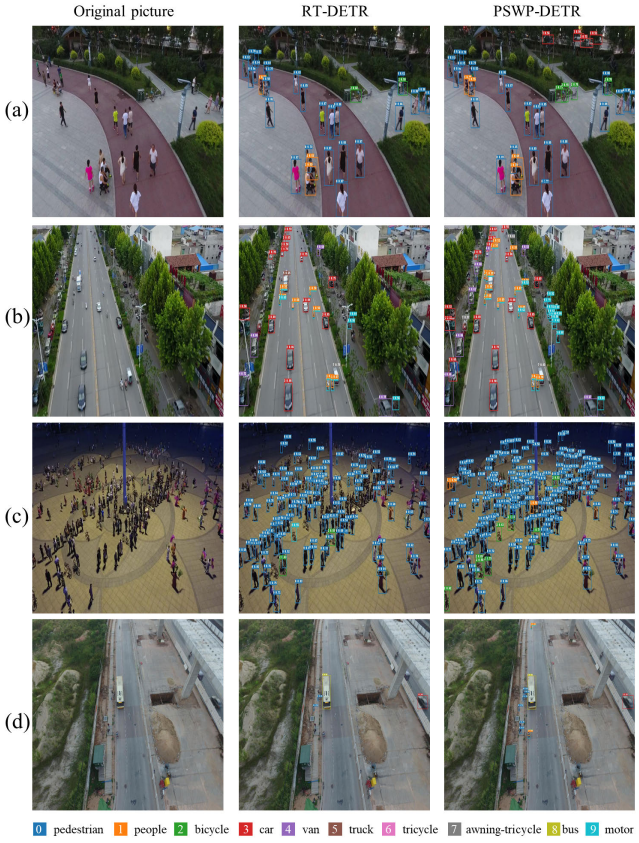


Fig. 9. Visual comparison of test results. (a) Objects in parks. (b) Objects in a traffic scene. (c) Pedestrians in a square. (c) Objects on a highway.

drops when any component was removed. For ② and ③, when deformation was unidirectional, reflect reduced accuracy due to RSIs' varied object shapes. Unidirectional deformation restricts feature capture across directions, affecting detection. For ④, without kernel deformation, fixed-size kernels might overlook or misalign with varied shapes. For ①, using all modules, PSWP-DETR through PdConv adaptively learns multidirectional deformation, better capturing diverse object textures and boosting accuracy.

2) Effectiveness Experiment of Modules in SDM: To verify the effectiveness of the modules within SDM, we first explored the scale sizes (s) of MSC, keeping all other

TABLE III
COMPARISON OF THE EFFECTIVENESS OF COMPONENTS IN PdConv

Method	AP^{val}	AP^{val}_{50}	AP^{val}_{75}	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}
① PSWP-DETR	32.5	54.1	32.5	22.3	45.4	64.1
② no PdConv_x	32.0	53.4	32.1	21.9	44.8	62.7
③ no PdConv_y	31.9	53.1	31.9	21.7	44.5	61.9
④ no Offset	32.1	53.4	32.3	22.0	44.9	63.2

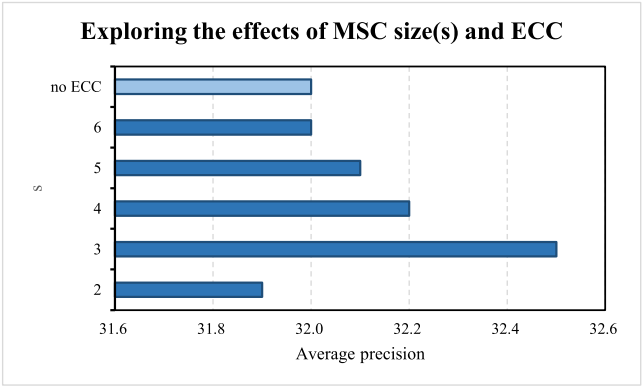


Fig. 10. Exploring the effects of MSC size(s) and ECC on average precision.

factors constant. We tested six different values for scales ranging from 2 to 6, corresponding to MSC sizes of {7, 5}, {7, 5, 3}, {7, 5, 3, 3}, {7, 5, 3, 3, 3}, and {7, 5, 3, 3, 3, 3}, respectively. Fig. 10 demonstrates the model's optimal performance when s is set to 3. Performance deteriorates when the value is less than or greater than this, due to excessive convolution leading to overfitting, and insufficient convolution failing to adequately extract features of remote sensing objects. Thus, $s = 3$ is identified as optimal for SDM. Subsequently, we explored the effectiveness of ECC. To facilitate observation, we consolidated the results into a single chart. We discovered that its absence leads to a decrease in AP. This is attributed to the removal of ECC diminishing the model's ability to extract an object's local features and edge information. Experiments demonstrate that each component is crucial.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

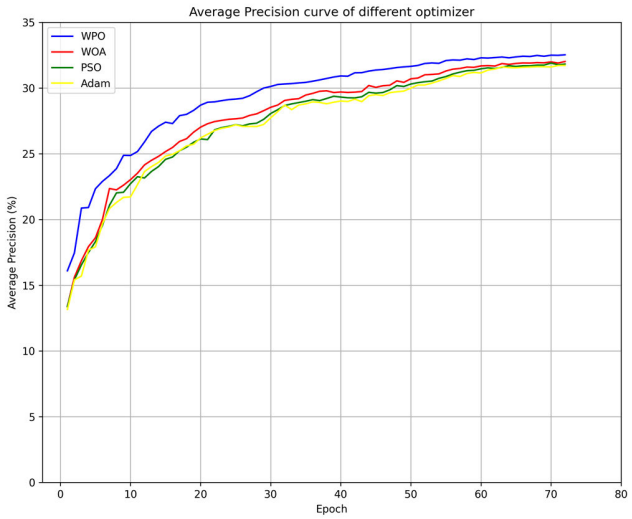


Fig. 11. Average precision of different optimizers.

3) Effectiveness Experiment of WPO: To evaluate the efficacy of the WPO algorithm, we assessed its performance on the PSWP-DETR model by comparing it with three other optimization algorithms: PSO, WOA, and Adam. This comparison utilized the VisDrone2019-DET dataset and controlled for all other variables to ensure a fair assessment of each algorithm's impact. In Fig. 11, we focused on how the mean average precision ($AP^{0.5:0.95}$) changes with training epochs to evaluate the performance of each optimization algorithm. The results demonstrated that WPO significantly outperformed the other algorithms. During the early stages of training, WPO quickly converged and achieved higher accuracy, whereas the performance of the other algorithms was relatively poor. PSO showed deficiencies in learning rate adjustment, resulting in a slower increase in AP values. Adam's performance was also suboptimal due to issues with task adaptability. Although WOA exhibited strong exploratory capabilities in the initial phases, its performance fluctuated significantly. In contrast, our improved WPO demonstrated higher stability. WPO successfully combines the advantages of PSO, where the algorithm's performance is influenced not only by each particle's current best position but also by the global best position recognized by the entire swarm. This combination allows WPO to avoid local optima more effectively, thereby finding more accurate optimal solutions. Compared to the other optimization algorithms, WPO's AP variation during model training was smoother, indicating a more stable learning process. This stability helps reduce the risk of significant performance differences across different datasets, effectively mitigating overfitting.

The experimental results further confirm that the WPO algorithm exhibits superior performance in RSOD tasks, demonstrating higher detection effectiveness.

E. Ablation Study

To validate the effectiveness of the proposed algorithm, we conducted the ablation study using PSWP-DETR on the VisDrone2019-DET dataset. Using the control variable method, we sequentially added PdConv, SDM, and WPO to the baseline model and combined them in various ways.

Table IV demonstrates that the model with PdConv alone achieved higher average precision than the baseline, with improvements of 1.3% in AP^{val}, 2.4% in APval 50, and 1.4% in APval 75. This improvement is due to PdConv's sophisticated feature extraction mechanism, which adaptively learns deformations to fit the diverse shapes of remote sensing targets. The model with SDM alone, when compared to the baseline, showed improvements of 1.6% in AP^{val}, 3.0% in APval 50, and 1.5% in APval 75, with a notable 2.0% increase in AP_S. This enhancement demonstrates the suitability of SDM for RSOD tasks that involve multiple scales, especially for detecting small-scale targets. The addition of PdConv and SDM has increased the model's parameters. Although this increase may limit deployment on resource-constrained devices, the performance improvement justifies this tradeoff in most practical applications. Combining PdConv and SDM with the baseline modeled to improvements of 2.3% in AP^{val}, 3.8% in AP^{val}_{50}, and 2.1% in AP^{val}_{75}, with a notable 2.4% increase in AP_S. This highlights their complementary role in feature extraction and scale adaptability, greatly enhancing the model's overall performance. Concerning learning rate optimization, WPO improves the detection accuracy, whether for the baseline model or any modified version. This enhancement is attributed to WPO's capability to dynamically adjust the learning rate in response to each model's performance during training, thereby optimizing the balance to facilitate accelerated convergence and enhanced overall performance. Ultimately, PSWP-DETR, which incorporates these three innovative methods, achieved the highest AP^{val} (+2.8%), AP^{val}_{50} (+4.6%), AP^{val}_{75} (+2.7%), AP^{val}_{S} (+3.1%), and AP^{val}_{M} (+2.8%). Although most methods showed a decrease in AP^{val}_{L}, the excellent mean precision achieved presents a high performance-to-cost ratio, especially for small target precision, which will be discussed in Section V. Despite the significant increase in computation and parameters, PSWP-DETR achieves a speed of 32.8 FPS. The experimental results comprehensively demonstrate the effectiveness of the various innovative modules on model detection performance.

F. Comparison of Popular Object Detectors

To further analyze the superiority of PSWP-DETR in RSOD, we utilized tools such as MMOLO [78] and detrex [79] to conduct experimental comparisons with 18 representative detectors, among which two [80], [81] are specifically tailored for RSOD, on the VisDrone2019-DET dataset. We broadly categorized these detectors into YOLO-series object detectors, transformer-based object detectors, and remote sensing object detectors. The experimental results are displayed in Table V.

The PSWP-DETR demonstrates superior performance compared to the YOLO-series object detectors, with a notable improvement in validation average precision (AP^{val}) ranging from 3.3% to 8.9%. This enhancement signifies a considerable advancement in the model's ability to detect objects accurately under various conditions. Generally, PSWP-DETR's inference speed is somewhat slower than that of most YOLO-series detectors. However, PSWP-DETR is notably 4.9 FPS faster than YOLOV6-L.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION EXPERIMENTS OF PSWP-DETR (THE OPTIMAL VALUES ARE BOLD, AND THE INCREMENTS ARE INDICATED IN THE BOTTOM-RIGHT CORNER OF THE RESPECTIVE NUMERICAL VALUES)

Method	AP^{val}	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params(M)
① RT-DETR	29.7	49.5	29.8	19.2	42.6	69.1	**43.25**	41.48
② + PdConv	$31.0_{+1.3}$	$51.9_{+2.4}$	$31.2_{+1.4}$	$21.0_{+1.8}$	$43.5_{+0.9}$	64.4	37.13	76.07
③ + SDM	$31.3_{+1.6}$	$52.5_{+3.0}$	$31.3_{+1.5}$	$21.2_{+2.0}$	$43.8_{+1.2}$	63.1	37.01	64.30
④ + WPO	$30.6_{+0.9}$	$51.3_{+1.8}$	$30.5_{+0.7}$	$20.2_{+1.0}$	$43.3_{+0.7}$	**70.2**$_{+1.1}$	43.14	41.48
⑤ + PdConv + SDM	$32.0_{+2.3}$	$53.3_{+3.8}$	$31.9_{+2.1}$	$21.6_{+2.4}$	$44.7_{+2.1}$	64.0	33.14	98.89
⑥ + PdConv + WPO	$31.6_{+1.9}$	$52.8_{+3.3}$	$31.7_{+1.9}$	$21.5_{+2.3}$	$44.3_{+1.7}$	65.3	37.07	76.07
⑦ + SDM + WPO	$31.8_{+2.1}$	$53.0_{+3.5}$	$32.1_{+\mathbf{2.3}}$	$21.7_{+2.5}$	$44.7_{+2.1}$	64.5	37.12	64.30
⑧ Ours	**32.5**$_{+2.8}$	**54.1**$_{+4.6}$	**32.5**$_{+2.7}$	**22.3**$_{+3.1}$	**45.4**$_{+2.8}$	64.1	32.8	98.89

In contrast to the transformer-based detection models, PSWP-DETR establishes a leading position by outperforming them in accuracy. Specifically, it achieves an improvement in AP^{val} ranging from 0.8% to an impressive 14.0% over its counterparts. Furthermore, PSWP-DETR excels in computational efficiency, maintaining lower giga floating-point operations per second (GFLOPs) while surpassing the DINO model in overall efficiency metrics. Detailed analysis reveals that PSWP-DETR not only requires fewer GFLOPs but also achieves a 0.8% higher AP^{val} compared to DINO.

The PSWP-DETR excels in most metrics among remote sensing detectors, highlighting its robust performance across various detection thresholds. Although it does not reach the highest precision in small object detection, it still maintains notable accuracy. Its detection of medium and large objects is particularly effective, underscoring the model's capability in handling large-scale detection tasks effectively and demonstrating its suitability for a wide range of professional remote sensing applications.

To visually assess each model's performance in the remote sensing dataset, we selected ten high-AP models from Table V to evaluate against PSWP-DETR on the VisDrone2019-DET test set, as shown in Fig. 12 (ID in Section IV-A1).

As shown in Fig. 12, while all detectors accurately identified some objects, they also produced false positives and missed some targets. For small targets captured at a greater imaging distance, such as the motor and person in the upper left corner of the image, models ①,③,④,⑤, and ⑥ failed to detect them. Models ②,⑦, and ⑧ were able to detect only a small portion, while model ⑨ performed reasonably well on small targets but misidentified a manhole cover as a truck.

Models ⑩ and ⑪ detected more small targets, with model ⑪ being more accurate in distinguishing pedestrians from people. For occluded cars that were closer, models ②,③,⑦, and ⑨ completely missed detection, while models ①,④,⑤,⑥, and ⑧ misidentified as trucks. Model ⑩ incorrectly classified the car and other objects as vans, while only model ⑪ correctly identified it as a car. Most models struggled to detect pixel-blurred objects, often overlooking bicycles and the people on them. Only model ⑪ successfully detected both the bicycle and the person simultaneously.

The visualization results indicate that PSWP-DETR outperforms these detectors in accurately identifying various remote sensing objects, particularly occluded objects and blurred objects with a lower pixel ratio due to being imaged from greater distances.

G. Comparison of Deformable Object Detectors

We compared PSWP-DETR with several popular deformable object detection methods, including Dq-YOLOF [82], FSTD-Net [83], Deformable YOLOv10L [25] [32], and Deformable DETR along with its variants, DAB-Deformable-DETR, and DN-Deformable-DETR [28], on the VisDrone2019-DET dataset. Deformable YOLOv10L is obtained by replacing some of the normal convolutions in YOLOv10L with deformable convolutions. The results of our experiments are presented in Table VI. PSWP-DETR outperforms all the other methods in terms of AP^{val}, AP^{val}_{50}, AP^{val}_{75}, AP^{val}_{S}, AP^{val}_{M}, and AP^{val}_{L}. Compared to the relatively high-precision DN-Deformable-DETR in the methods, although PSWP-DETR has twice the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

TABLE V
COMPARISON EXPERIMENT ON VISDRONE2019-DET (THE HIGHEST VALUES IN THE TABLE ARE MARKED IN RED, FOLLOWED BY BLUE FOR THE SECOND HIGHEST)

Method	Backbone	Epoch	Params(M)	GFLOPs(G)	FPS	AP^{val}	$AP^{val}_{0.5}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}
YOLO-series Object Detectors											
YOLOX-L	–	140	54.21	77.83	33.6	23.8	36.0	25.8	14.0	35.2	46.0
YOLOv6-L	–	140	58.47	71.33	27.9	25.6	37.7	28.3	14.0	39.8	55.5
YOLOv7-L	–	140	36.55	51.82	42.5	24.8	37.4	26.9	14.2	37.4	45.9
YOLOv7-X	–	140	70.88	94.31	38.9	26.1	39.3	28.3	15.4	38.8	50.7
YOLOv8-L	–	140	43.64	82.57	42.4	23.6	33.2	26.7	12.2	37.5	53.5
YOLOv8-X	–	140	68.16	129	40.2	25.1	35.3	28.1	13.2	39.5	54.8
YOLOV9-C [24]	–	130	50.72	236.7	41.4	25.0	41.7	25.2	15.1	38.1	51.3
YOLOv9-E [24]	–	130	68.56	240.8	35.7	29.2	47.0	30.4	19.9	42.5	57.1
YOLOv10-L [25]	–	130	25.73	126.4	**74.6**	24.9	41.0	25.4	15.1	37.4	45.3
YOLOv10-X [25]	–	130	31.6	169.9	**54.0**	25.4	42.3	25.8	15.8	37.9	43.3
Transformer-based Object Detectors											
DN-DETR-DC5	Resnet 50	50	43.7	199.5	–	20.3	38.5	18.7	11.4	30.7	43.2
DAB-DETR-DC5		50	43.7	199.5	–	21.3	39.5	20.5	12.1	32.2	45.6
Anchor-DETR-DC5		50	37.02	194.7	–	21.8	39.8	20.7	12.6	33.0	51.2
Focus-DETR		12	49.20	171.5	–	31.1	53.3	31.0	21.8	43.0	57.5
DINO		12	47.51	273.7	–	**31.7**	**53.8**	**31.5**	**22.6**	**43.1**	60.3
RT-DETR		72	41.48	68.19	42.6	29.7	49.5	29.8	19.2	42.6	**69.1**
Remote Sensing Object Detectors											
Ma et al. [80]	–	120	–	–	–	30.2	52.5	30.6	**22.6**	39.6	39.8
DTSSNet [81]	MobileNetV2	50	10.1	–	–	24.2	39.9	25.2	17.5	32.3	36.1
PSWP-DETR (ours)	Resnet 50	72	98.89	140.86	32.8	**32.5**	**54.1**	**32.5**	**22.3**	**45.4**	**64.1**

number of parameters, its GFLOPS is approximately half that of DN-Deformable-DETR. In addition, PSWP-DETR demonstrates better performance across several metrics, including AP^{val} (+2.5%), AP^{val}_{50} (+2.1%), AP^{val}_{75} (+3.2%), AP^{val}_{S} (+0.8%), AP^{val}_{M} (+4.2%), and AP^{val}_{L} (+9.0%). This indicates that PSWP-DETR delivers superior detection performance for targets in RSIs among these deformable methods.

H. Comparison of Public Datasets

To evaluate the effectiveness of PSWP-DETR, we employed the DIOR and AI-TOD datasets, adhering to the setup outlined in Section IV-B, utilizing 640 × 640 inputs for 72 epochs. Fig. 13 illustrates the AP performance of the models, while Table VII presents a comprehensive comparison of the results obtained by PSWP-DETR and RT-DETR on these datasets. PSWP-DETR demonstrated superior performance over RT-DETR, achieving increases in all precision metrics. This underscores its proficiency in detecting remote sensing objects, particularly on the DIOR dataset. In comparison to RT-DETR, PSWP-DETR exhibited slight improvements in AP^{val} on AI-TOD. Figs. 14 and 15 demonstrate the detection capabilities of RT-DETR and PSWP-DETR on the DIOR and AI-TOD datasets, respectively. In Fig. 14, RT-DETR failed to detect smaller airplanes and could not delineate airport boundaries (a), overlooked numerous harbors (b), and ignored small vehicles (c). Conversely, PSWP-DETR excelled at identifying diverse objects, such as airplanes, harbors, and vehicles, even with minimal pixel coverage, owing to its superior feature extraction and scale integration capabilities. In Fig. 15, RT-DETR missed many vehicles in a nighttime parking scenario (a), whereas PSWP-DETR accurately identified more vehicles, demonstrating superior low-light performance.

In groups (b) and (c), while both detectors identified numerous small objects, PSWP-DETR detected a greater number and more accurately overall.

The experimental results indicate that our improved PSWP-DETR demonstrates enhanced capabilities in accurately identifying diverse objects across various scenarios, owing to its superior feature extraction and scale integration capabilities.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON WITH SOME DEFORMATION-LEARNING METHODS (THE BETTER VALUES IN THE TABLE WILL BE BOLDED)

Method	Params(M)	GFLOPs(G)	AP^{val}	$AP^{val}_{0.5}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}
Dq-YOLOF	-	-	21.8	38.5	23.7	13.5	34.7	39.2
FSTD-Net	-	-	23.2	41.5	23.0	13.2	35.1	42.1
Deformable YOLOv10L	25.71	493.5	28.8	47.2	29.8	20.5	40.0	39.6
Deformable-DETR	40.05	193	18.5	34.1	17.3	10.2	28.2	45.1
DAB-Deformable-DETR	47.3	260	29.2	51.7	28.3	20.6	40.8	52.8
DN-Deformable-DETR	47.3	260	30.0	52.0	29.3	21.5	41.2	53.1
PSWP-DETR	98.89	140.86	**32.5**	**54.1**	**32.5**	**22.3**	**45.4**	**64.1**

TABLE VII
VALIDATION ON PUBLIC DATASETS

Datasets	Method	Params(M)	GFLOPs(G)	FPS	AP^{val}	AP^{val}_{50}	AP^{val}_{75}	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}
DIOR	RT-DETR	41.48	68.21	59.72	57.9	79.6	62.5	22.8	50.1	75.7
	PSWP-DETR	98.90	140.88	35.32	62.2	83.5	67.5	25.1	54.3	80.3
AI-TOD	RT-DETR	41.48	68.19	58.70	24.0	54.8	17.4	22.9	43.5	–
	PSWP-DETR	98.89	140.85	35.84	24.4	56.1	17.6	23.6	40.5	–

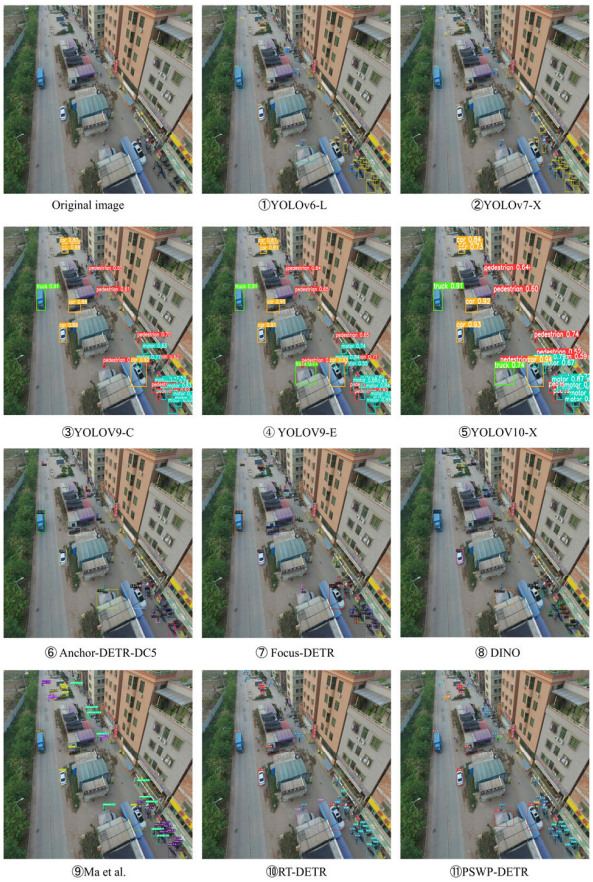


Fig. 12. Visualization of comparison experiments on the VisDrone2019-DET dataset using selected detectors.

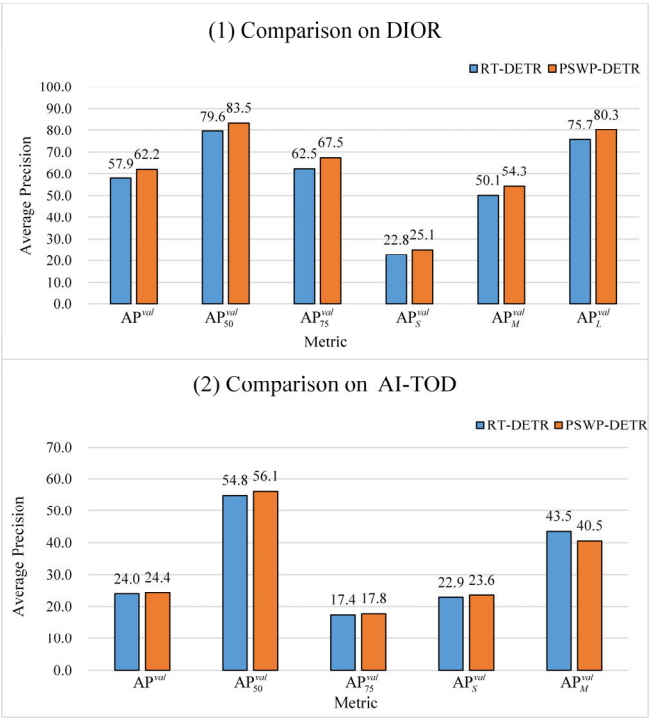


Fig. 13. Comparison of public datasets. (a) Comparison of DIOR. (b) Comparison of AI-TOD.

V. SHORTCOMINGS AND DISCUSSION

The experimental results show that PSWP-DETR has superior detection for variously shaped and scaled objects in RSIs, excelling at identifying small objects. Notably, PSWP-DETR surpasses previous transformer models, offering superior RSOD with less computational need and faster speed. However, despite its strong performance, some challenges remain. This section explores drawbacks and suggests improvements.

1) Table II demonstrates that PSWP-DETR excels in average precision and recall on the VisDrone2019-DET dataset. However, its complex design increases the number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

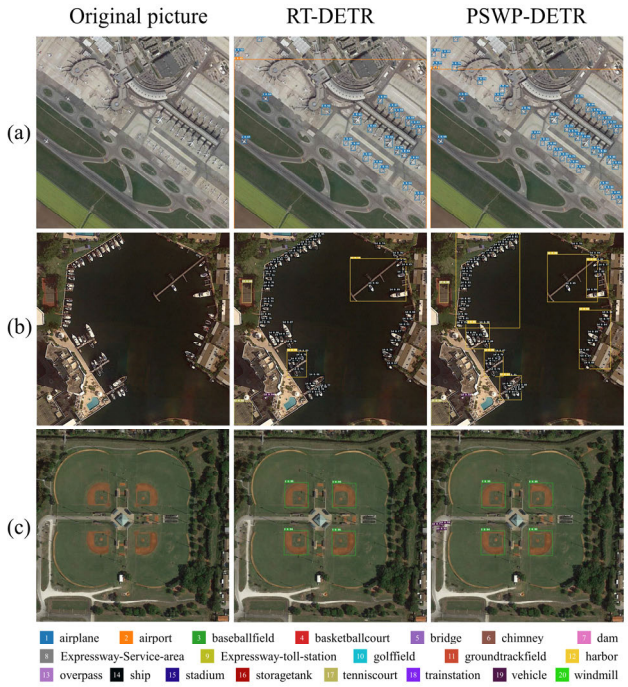


Fig. 14. Visual comparison experiment on DIOR. (a) Objects in the airport scene. (b) Objects in the harbor scene. (c) Objects in a living scene.

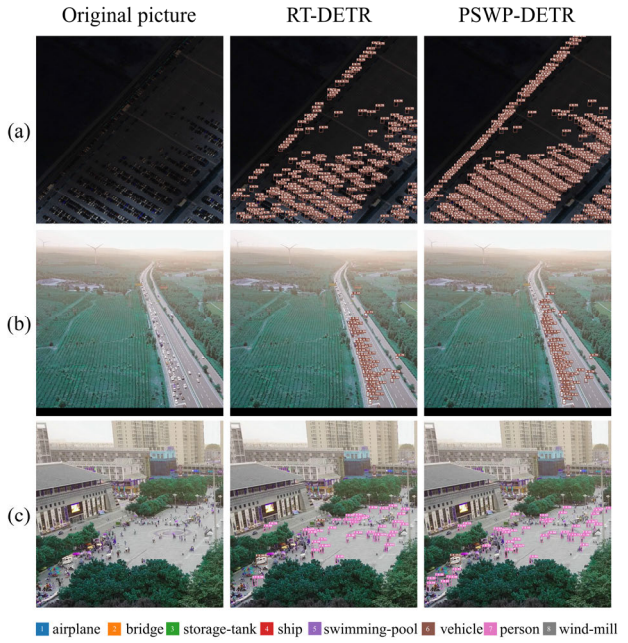


Fig. 15. Visual comparison experiment on AI-TOD. (a) Parking lots in a dark environment. (b) Vehicles on the highway. (c) Objects in a square.

of parameters and computational demands, which poses challenges in resource-constrained environments. While PSWP-DETR achieves 32.8 FPS, it is 9.8 FPS slower than RT-DETR. This is attributed to the complexity of feature extraction by PdConv and SDM. A key challenge in advancing the model's real-time applications is how to significantly improve the accuracy of RSOD while simultaneously reducing computational demands. Future research will focus on developing more lightweight methods for RSOD [84].

2) For the DIOR dataset, which includes remote sensing objects with significant scale and shape differences, PSWP-DETR demonstrates superior detection performance compared to RT-DETR (+4.3% AP, +3.9% AP^{50}, +5.0% AP^{75}, +2.3% AP_S, +4.2% AP_M, and +4.6% AP_L). However, for categories with fewer samples, there may be issues with insufficient feature learning. For example, as shown in Fig. 15, group (b), PSWP-DETR failed to detect larger windmills. This oversight is attributed to the AI-TOD dataset's emphasis on smaller objects and scarcity of larger ones, also mirroring the cause of its reduced AP_L^{val} on the VisDrone2019-DET dataset, as indicated in Table II. This imbalance in the datasets may have constrained the model's ability to learn features necessary for accurately detecting larger objects. Enhancing the model's ability to learn underrepresented categories is an important direction for future research [85], [86].

VI. Conclusion

Variability in the shape and scale of objects in RSIs complicates detection. In this work, we developed PSWP-DETR, a transformer-based remote sensing object detector. First, we proposed PredatorConv (PdConv) adaptively learning deformations based on the input feature map, to closely fit the objects' shapes, thereby perceiving the features of variably shaped objects in RSIs. Then, we introduced SDMs to merge multiscale features with edge details. SDM uses MSC for blending multiscale attributes and ECC to capture edge information and local textures, easing the challenges of detecting multiscale and especially small objects amid imaging and complex backgrounds. Finally, we proposed a WPO algorithm to find the optimal learning rate and enhance the model's detection performance. PSWP-DETR achieves 54.1% (+4.6%) AP^{50} on VisDrone2019-DET, 83.5% (+3.9%) AP^{50} on DIOR, and 56.1% (+1.3%) AP^{50} on AI-TOD. Compared to existing remote sensing object detectors, PSWP-DETR demonstrates superior accuracy in detecting objects of various scales and shapes, particularly small targets with blurred edges and fewer pixels. This highlights the model's potential for advancing future RSOD efforts. To address the potential limitations of PSWP-DETR, future improvements will focus on two key areas: reducing the model's computational complexity while maintaining high-precision detection of remote sensing objects and enhancing the model's ability to learn from underrepresented categories in imbalanced datasets.

References

[1] A. Lin et al., "Identifying urban building function by integrating remote sensing imagery and POI data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8864–8875, 2021, doi: 10.1109/JSTARS.2021.3107543.

[2] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[3] Y. Wang et al., "Remote sensing image super-resolution and object detection: Benchmark and state of the art," *Expert Syst. Appl.*, vol. 197, Jul. 2022, Art. no. 116793.

[4] W. Qiao, L. Shen, Q. Wen, Q. Wen, S. Tang, and Z. Li, "Revolutionizing building damage detection: A novel weakly supervised approach using high-resolution remote sensing images," *Int. J. Digit. Earth*, vol. 17, no. 1, Dec. 2024, Art. no. 2298245.

[5] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

[6] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[7] G. Liu, X. Sun, K. Fu, and H. Wang, "Aircraft recognition in high-resolution satellite images using coarse-to-fine shape prior," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 3, pp. 573–577, May 2013.

[8] Z. Shi, X. Yu, Z. Jiang, and B. Li, "Ship detection in high-resolution optical imagery based on anomaly detector and local shape feature," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4511–4523, Aug. 2014.

[9] M. Gong, S. Zhao, L. Jiao, D. Tian, and S. Wang, "A novel coarse-to-fine scheme for automatic image registration based on SIFT and mutual information," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4328–4338, Jul. 2014.

[10] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412.

[11] D. Hong, J. Yao, C. Li, D. Meng, N. Yokoya, and J. Chanussot, "Decoupled-and-coupled networks: Self-supervised hyperspectral image super-resolution with subpixel fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5527812.

[12] J. Zhan et al., "DGPF-RENet: A low data dependence network with low training iterations for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520521.

[13] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-Net in U-Net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.

[14] P. Shamsolmoali, M. Zareapoor, J. Chanussot, H. Zhou, and J. Yang, "Rotation equivariant feature image pyramid network for object detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5608614.

[15] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215.

[16] Y. Shen, D. Liu, F. Zhang, and Q. Zhang, "Fast and accurate multi-class geospatial object detection with large-size remote sensing imagery using CNN and truncated NMS," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 235–249, Sep. 2022.

[17] Z. Chen et al., "TO–YOLOX: A pure CNN tiny object detection model for remote sensing images," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 3882–3904, Oct. 2023, doi: 10.1080/17538947.2023.2261901.

[18] X. Xie, G. Cheng, Q. Li, S. Miao, K. Li, and J. Han, "Fewer is more: Efficient object detection in large aerial images," *Sci. China Inf. Sci.*, vol. 66, pp. 1–18, Aug. 2023, doi: 10.1007/s11432-022-3718-5.

[19] A. S. M. S. Sagar, Y. Chen, Y. Xie, and H. S. Kim, "MSA R-CNN: A comprehensive approach to remote sensing object detection and scene understanding," *Expert Syst. Appl.*, vol. 241, Nov. 2023, Art. no. 122788, doi: 10.1016/j.eswa.2023.122788.

[20] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[21] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[22] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2016.

[23] J. Terven and D. Cordova-Esparza, "A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS," 2023, *arXiv:2304.00501*.

[24] C.-Y. Wang, I.-H. Yeh, and H.-Y. Mark Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," 2024, *arXiv:2402.13616*.

[25] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.

[26] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[27] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*, Jan. 2020, pp. 213–229.

[28] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. ICLR*, 2021, pp. 1–9.

[29] Y. Li, N. Miao, L. Ma, F. Shuang, and X. Huang, "Transformer for object detection: Review and benchmark," *Eng. Appl. Artif. Intell.*, vol. 126, Nov. 2023, Art. no. 107021.

[30] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[31] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.

[32] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[33] Z. Deng, H. Sun, S. Zhou, J. Zhao, L. Lei, and H. Zou, "Multi-scale object detection in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 3–22, Nov. 2018.

[34] Z. Liu et al., "A hierarchical progressive recognition network for building change detection in high-resolution remote sensing images," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 40, no. 2, pp. 243–262, Jan. 2025.

[35] X. Xu, J. Li, and Z. Chen, "TCIANet: Transformer-based context information aggregation network for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1951–1971, 2023.

[36] Z. Lv, P. Zhang, W. Sun, J. A. Benediktsson, J. Li, and W. Wang, "Novel adaptive region spectral–spatial features for land cover classification with high spatial resolution remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609412.

[37] Y. Zhou, S. Chen, J. Zhao, R. Yao, Y. Xue, and A. E. Saddik, "CLT-det: Correlation learning based on transformer for detecting dense objects in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4708915.

[38] J. Xue, D. He, M. Liu, and Q. Shi, "Dual network structure with interweaved global-local feature hierarchy for transformer-based object detection in remote sensing image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6856–6866, 2022.

[39] M. Li et al., "Remote sensing object detection based on strong feature extraction and prescreening network," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[40] L. Bottou, "Stochastic gradient descent tricks," in *Neural Networks: Tricks of the Trade*. Berlin, Germany: Springer, 2012, pp. 421–436.

[41] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[42] J. Duchi, E. Hazan, and Y. Singer, "Adaptive subgradient methods for online learning and stochastic optimization," *J. Mach. Learn. Res.*, vol. 12, no. 7, pp. 2121–2159, Jul. 2011.

[43] A. C. Wilson, R. Roelofs, M. Stern, N. Srebro, and B. Recht, "The marginal value of adaptive gradient methods in machine learning," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2017, pp. 4148–4158.

[44] M. Ragab, "Leveraging mayfly optimization with deep learning for secure remote sensing scene image classification," *Comput. Electr. Eng.*, vol. 108, May 2023, Art. no. 108672.

[45] S. Huang, C. Lin, X. Jiang, and Z. Qu, "BRSTD: Bio-inspired remote sensing tiny object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5643115.

[46] X. Guo, B. Hou, B. Ren, Z. Ren, and L. Jiao, "Network pruning for remote sensing images classification based on interpretable CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605615.

[47] Y. Lu, M. Gong, Z. Hu, W. Zhao, Z. Guan, and M. Zhang, "Energy-based CNNs pruning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3000214.

[48] C. Li, G. Cheng, G. Wang, P. Zhou, and J. Han, "Instance-aware distillation for efficient object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602011.

[49] P. Li et al., "Hashing nets for hashing: A quantized deep learning to hash framework for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7331–7345, Oct. 2020.

[50] J. Zhan et al., "Heterogeneous binary pixel difference networks for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5602619, doi: 10.1109/TGRS.2024.3520161.

[51] J. Zhang, J. Lei, W. Xie, Y. Li, G. Yang, and X. Jia, "Guided hybrid quantization for object detection in remote sensing imagery via one-to-one self-teaching," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5614815.

[52] X. Zhu, S. Lyu, X. Wang, and Q. Zhao, "TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 2778–2788.

[53] J. Lin, Y. Zhao, S. Wang, and Y. Tang, "YOLO-DA: An efficient YOLO-based detector for remote sensing object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.

[54] A. Wang, Y. Xu, H. Wang, Z. Wu, and Z. Wei, "CDE-DETR: A real-time end-to-end high-resolution remote sensing object detection method based on RT-DETR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2024, pp. 8090–8094.

[55] H. Kim, S. Joung, I.-J. Kim, and K. Sohn, "Shape-adaptive kernel network for dense object detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 2046–2050.

[56] F. Chen, F. Wu, J. Xu, G. Gao, Q. Ge, and X.-Y. Jing, "Adaptive deformable convolutional network," *Neurocomputing*, vol. 453, pp. 853–864, Sep. 2021.

[57] R. Jing, W. Zhang, Y. Li, W. Li, and Y. Liu, "Feature aggregation network for small object detection," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124686.

[58] D. Yu and S. Ji, "A new spatial-oriented object detection framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Nov. 2022, Art. no. 4407416, doi: 10.1109/TGRS.2021.3127232.

[59] Y. Qi, Y. He, X. Qi, Y. Zhang, and G. Yang, "Dynamic snake convolution based on topological geometric constraints for tubular structure segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6047–6056.

[60] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.

[61] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9196–9205.

[62] G. Ghiasi, T.-Y. Lin, and Q. V. Le, "NAS-FPN: Learning scalable feature pyramid architecture for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7029–7038.

[63] A. Xu, A. Yao, A. Li, A. Liang, and A. Zhang, "Auto-FPN: Automatic network architecture adaptation for object detection beyond classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6648–6657.

[64] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.

[65] Y. Chen, X. Yuan, R. Wu, J. Wang, Q. Hou, and M.-M. Cheng, "YOLO-MS: Rethinking multi-scale representation learning for real-time object detection," 2023, *arXiv:2308.05480*.

[66] T. Gao, Z. Li, Y. Wen, T. Chen, Q. Niu, and Z. Liu, "Attention-free global multiscale fusion network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603214, doi: 10.1109/TGRS.2023.3346041.

[67] D. Wang, D. Tan, and L. Liu, "Particle swarm optimization algorithm: An overview," *Soft Comput.*, vol. 22, no. 2, pp. 387–408, Jan. 2018.

[68] C. Blum, "Ant colony optimization: Introduction and recent trends," *Phys. Life Rev.*, vol. 2, no. 4, pp. 353–373, Dec. 2005.

[69] J. Xue and B. Shen, "A novel swarm intelligence optimization approach: Sparrow search algorithm," *Syst. Sci. Control Eng.*, vol. 8, no. 1, pp. 22–34, Jan. 2020.

[70] S. Mirjalili, S. M. Mirjalili, and A. Lewis, "Grey wolf optimizer," *Adv. Eng. Softw.*, vol. 69, pp. 46–61, Jan. 2014.

[71] S. Mirjalili and A. Lewis, "The whale optimization algorithm," *Adv. Eng. Softw.*, vol. 95, no. 8, pp. 51–67, May 2016.

[72] W. Dai et al., "AISOA-SSformer: An effective image segmentation method for rice leaf disease based on the transformer architecture," *Plant Phenomics*, vol. 6, p. 218, Aug. 2024.

[73] A. H. Khan, X. Cao, S. Li, V. N. Katsikis, and L. Liao, "BAS-ADAM: An Adam based approach to improve the performance of beetle antennae search optimizer," *IEEE/CAA J. Autom. Sinica*, vol. 7, no. 2, pp. 461–471, Mar. 2020.

[74] D. Du et al., "VisDrone-DET2019: The vision meets drone object detection in image challenge results," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Seoul, South Korea, Oct. 2019, pp. 213–226.

[75] J. Wang, W. Yang, H. Guo, R. Zhang, and G.-S. Xia, "Tiny object detection in aerial images," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 3791–3798.

[76] Z. Yu et al., "Searching central difference convolutional networks for face anti-spoofing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5294–5304.

[77] Y. Ma, D. Yu, T. Wu, and H. Wang, "PaddlePaddle: An open-source deep learning platform from industrial practice," *Frontiers Data Domputing*, vol. 1, no. 1, pp. 105–115, 2019.

[78] MMYOLO Contributors. (2022). *MMYOLO: OpenMMLab YOLO Series Toolbox and Benchmark*. [Online]. Available: https://github.com/open-mmlab/mmyolo

[79] T. Ren et al., "Detrex: Benchmarking detection transformers," 2023, *arXiv:2306.07265*.

[80] Y. Ma, L. Chai, and L. Jin, "Scale decoupled pyramid for object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4704314.

[81] L. Chen, C. Liu, W. Li, Q. Xu, and H. Deng, "DTSSNet: Dynamic training sample selection network for UAV object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5902516, doi: 10.1109/TGRS.2023.3348555.

[82] X. Qi, M. G. Md Johar, A. Khatibi, J. Tham, and L. Cheng, "Dq-YOLOF: An effective improvement with deformable convolution and sample quality optimization based on the YOLOF detector," *Electronics*, vol. 13, no. 21, p. 4204, Oct. 2024.

[83] W. Yu, J. Zhang, D. Liu, Y. Xi, and Y. Wu, "An effective and lightweight full-scale target detection network for UAV images based on deformable convolutions and multi-scale contextual feature optimization," *Remote Sens.*, vol. 16, no. 16, p. 2944, Aug. 2024.

[84] Y. Wang et al., "RingMo-lite: A remote sensing lightweight network with CNN-transformer hybrid framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5608420.

[85] W. Zhang and Y.-X. Wang, "Hallucination improves few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13003–13012.

[86] G. Zhang, Z. Luo, K. Cui, S. Lu, and E. P. Xing, "Meta-DETR: Image-level few-shot detection with inter-class correlation exploitation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 12832–12843, Nov. 2023, doi: 10.1109/TPAMI.2022.3195735.



Xiyu Zhong is currently pursuing with the Central South University of Forestry and Technology, Changsha, China.

Her main research interests include deep learning and remote sensing image processing.



Jialei Zhan is currently pursuing with the National University of Defense Technology, Changsha, China.

His main research interests include computer vision and remote sensing image processing.



Yuhang Xie is currently pursuing with Xidian University, Xi'an, China.

His main research interests include deep learning, and graphics and image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.



Lingtao Zhang is currently an Associate Professor with the Central South University of Forestry and Technology, Changsha, China. His research interests include pattern recognition, big data analytics, and machine learning.



Guoxiong Zhou received the Ph.D. degree from Central South University, Changsha, China, in 2010.

He is currently a Professor with the Central South University of Forestry and Technology, Changsha. His main research interests include forest fire prevention and robotics.



Mingyue Liang is currently pursuing with the Central South University of Forestry and Technology, Changsha, China.

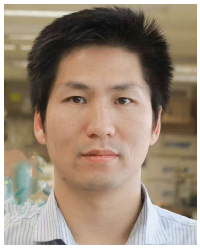
Her main research interests include data mining and deep learning.



Kaitai Yang is currently pursuing with the Central South University of Forestry and Technology, Changsha, China. His main research interests include pattern recognition and deep learning.



Zonghao Guo is currently pursuing with the Central South University of Forestry and Technology, Changsha, China. His main research interests include computer vision and deep learning.



Liujun Li is currently an Associate Research Professor with Missouri University of Science and Technology, Rolla, MO, USA. His research interests include robotics sensing, control, computing, and AI applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:11:04 UTC from IEEE Xplore. Restrictions apply.